The New Classical Approach

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1 Introduction

The New Classical approach extends monetarist skepticism about policy fine-tuning by sharpening the mechanism through which policy affects real outcomes. The key shift is from short-run money illusion under adaptive expectations (Friedman's framework) to model-consistent forecasting under rational expectations, which removes systematic policy surprises and eliminates systematic real effects of monetary policy.

1.1 Expectations and the rational expectations revolution

1.1.1 Adaptive expectations

Adaptive expectations operate as a trial-and-error forecasting rule where agents update forecasts using past observed outcomes:

Price expectations: $E_{t-1}P_t = f(P_{t-1}, P_{t-2}, ...)$

A common error-correction form for expected inflation is:

$$\pi_t^e = \pi_{t-1}^e + \lambda(\pi_{t-1} - \pi_{t-1}^e), \quad 0 < \lambda \leq 1.$$

Here, π_t^e is expected inflation in period t, π_{t-1} is actual inflation in period $t-1$, and λ measures how quickly agents revise their forecast after making an error.

Adaptive expectations permit systematic forecast errors during regime changes, creating temporary money illusion and thus temporary real effects.

1.1.2 Rational expectations

Rational expectations are defined by three criteria: (i) forecasts use the true economic model (agents know the economy's structure), (ii) forecasts condition on all available information, and (iii) forecast errors are purely random—systematic errors are eliminated.

In classical formulation: expectations are "the same as the predictions of the relevant economic theory," and the economy does not waste information.

Model-consistent forecasting:

$$E_{t-1}P_t = E(P_t \mid I_{t-1}),$$

where I_{t-1} is the information set at time $t-1$. In words: the expectation formed in period $t-1$ is the best model-consistent forecast of the price level in period t, using all information available at the time the forecast is made.

If workers have access to unbiased forecasts, predictable policy loses its ability to manipulate perceived real wages and thus loses its ability to shift real output, even in the short run.

2 The New Classical model of output and prices

2.1 Lucas aggregate supply

The New Classical model begins with a supply relationship in which output depends on unexpected movements in the price level, not on predictable changes that agents have already incorporated into their decisions.

The Lucas aggregate supply (LAS) curve is therefore upward sloping in (Y, P) space: when the actual price level turns out higher than expected, firms temporarily expand output; when it turns out lower than expected, output falls below potential. The figure below illustrates this benchmark relationship.

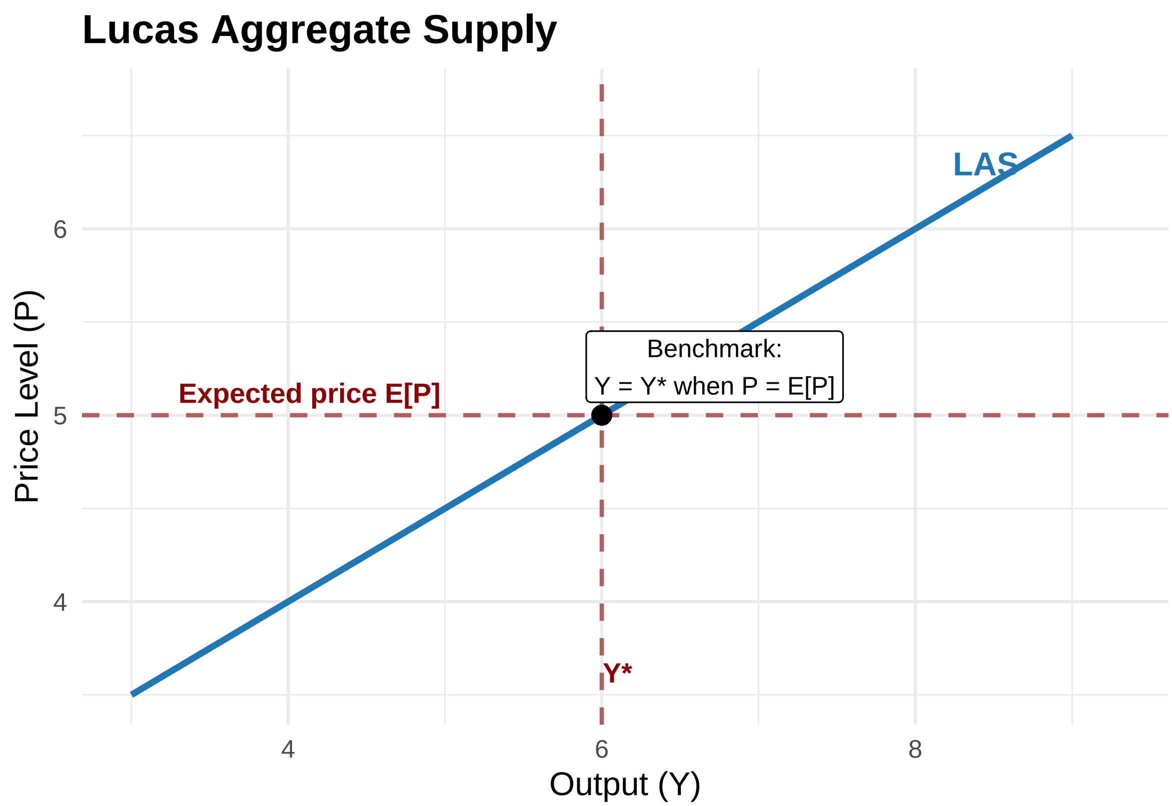

Figure 1: Lucas aggregate supply curve. The benchmark point is where the actual price level equals the expected price level, so output remains at potential Y^*. Output rises above Y^* only when the price level is unexpectedly high, and falls below Y^* when the price level is unexpectedly low.

In the figure, the black point marks the benchmark equilibrium on the supply side where expectations are fulfilled: $P_t = E_{t-1}P_t$ and therefore $Y_t = Y^*$.

Lucas aggregate supply (LAS): Output deviates from potential only when the actual price level differs from the price level that firms and workers expected one period earlier:

$$Y_t^s - Y^* = \kappa(P_t - E_{t-1}P_t).$$

Lucas (1972) derives this in a market-clearing framework with optimizing agents.

Each term has a clear economic meaning: - Y_t^s is actual output supplied in period t - Y^* is potential or natural output - $P_t - E_{t-1}P_t$ is the unexpected part of the price level - κ measures how strongly output responds to that price surprise

Interpretation in the diagram: - If $P_t > E_{t-1}P_t$: the economy moves up along the LAS curve and output rises above potential ($Y_t^s > Y^*$) - If $P_t = E_{t-1}P_t$: the economy stays at the benchmark point, so output remains at potential ($Y_t^s = Y^*$) - If $P_t < E_{t-1}P_t$: the economy moves down along the LAS curve and output falls below potential ($Y_t^s < Y^*$)

The parameter κ measures supply responsiveness: if the actual price level exceeds the expected price level by one unit, output rises by κ units. In calculus notation, $\partial Y_t / \partial(P_t - E_{t-1}P_t) = \kappa$.

2.2 Aggregate demand

To complete the New Classical model, aggregate demand links output to nominal money and the price level.

Aggregate demand (AD): In log form:

$$Y_t^d = \gamma + M_t - P_t,$$

where M_t is nominal money and γ captures exogenous demand shifts.

This is a reduced-form aggregate demand equation. Since $M_t - P_t$ is the log of real money balances, the equation says that demand is higher when the real money stock is higher. The parameter γ collects other influences on demand, such as velocity or autonomous spending.

Agents know structural parameters γ, κ, and natural output Y^*.

2.3 Solving the model: the role of κ

Equilibrium requires supply to equal demand:

$$Y_t^s = Y_t^d = Y_t.$$

Substitute the two equations into equilibrium:

$$Y_t - Y^* = \kappa(P_t - E_{t-1}P_t),$$

$$Y_t = \gamma + M_t - P_t.$$

The logic is easiest to see in three steps.

1. Under rational expectations:

$$E_{t-1}Y_t = Y^*.$$

Why? Taking expectations of the LAS equation one period earlier gives:

$$E_{t-1}(Y_t - Y^*) = \kappa E_{t-1}(P_t - E_{t-1}P_t).$$

Because $E_{t-1}P_t$ is already known at time $t-1$, the expected price surprise is zero:

$$E_{t-1}(P_t - E_{t-1}P_t) = 0.$$

So:

$$E_{t-1}Y_t = Y^*.$$

2. Output deviations depend only on monetary surprises:

$$Y_t - Y^* = \frac{\kappa}{1+\kappa}(M_t - E_{t-1}M_t).$$

To derive this, first use AD to write price as:

$$P_t = \gamma + M_t - Y_t.$$

Substitute this into LAS:

$$Y_t - Y^* = \kappa[(\gamma + M_t - Y_t) - E_{t-1}P_t].$$

Now take expectations at time $t-1$ using $E_{t-1}Y_t = Y^*$:

$$Y^* = \gamma + E_{t-1}M_t - E_{t-1}P_t.$$

Rearranging gives:

$$E_{t-1}P_t = \gamma + E_{t-1}M_t - Y^*.$$

Substitute this back into the LAS equation:

$$Y_t - Y^* = \kappa[(\gamma + M_t - Y_t) - (\gamma + E_{t-1}M_t - Y^*)].$$

Simplifying:

$$Y_t - Y^* = \kappa[(M_t - E_{t-1}M_t) - (Y_t - Y^*)].$$

Collect terms in $Y_t - Y^*$:

$$(1+\kappa)(Y_t - Y^*) = \kappa(M_t - E_{t-1}M_t).$$

Hence:

$$Y_t - Y^* = \frac{\kappa}{1+\kappa}(M_t - E_{t-1}M_t).$$

3. Price surprises scale with monetary surprises:

$$P_t - E_{t-1}P_t = \frac{1}{1+\kappa}(M_t - E_{t-1}M_t).$$

This follows directly from the LAS equation. Divide the output result in step 2 by κ:

$$P_t - E_{t-1}P_t = \frac{Y_t - Y^*}{\kappa} = \frac{1}{1+\kappa}(M_t - E_{t-1}M_t).$$

Interpretation: larger κ means output responds more, prices respond less; smaller κ reverses this. The Lucas supply slope (P on vertical axis, Y horizontal) is proportional to $1/\kappa$.

When policy is volatile, agents discount price movements as noise ("boy who cried wolf"), lowering effective κ and steepening LAS.

3 Monetary policy implications

3.1 The monetary policy ineffectiveness proposition

With rational expectations and market clearing, *systematic and predictable* monetary policy has **no real effect**, even short-run. Predictable policy is incorporated into expectations, shifting prices, not real outcomes.

Formally, if $M_t = E_{t-1}M_t$ (policy is forecastable):

$$Y_t - Y^* = 0.$$

3.2 Systematic rules versus random disturbances

The key distinction in the New Classical model is between a policy rule that agents can anticipate and a monetary disturbance that arrives as a surprise. If policy is predictable, expectations adjust in advance and output stays at potential.

The figure contrasts those two cases. The red AD curve represents the predictable-policy case, where equilibrium remains at Y^*. The green AD curve represents an unexpected monetary expansion, which temporarily shifts demand and moves output above potential.

The figure makes the contrast explicit: predictable policy leaves the economy at Y^*, whereas an unexpected disturbance shifts AD and temporarily pushes output away from potential.

Money supply process:

$$M_t = (1+g_M)M_{t-1} + u_t,$$

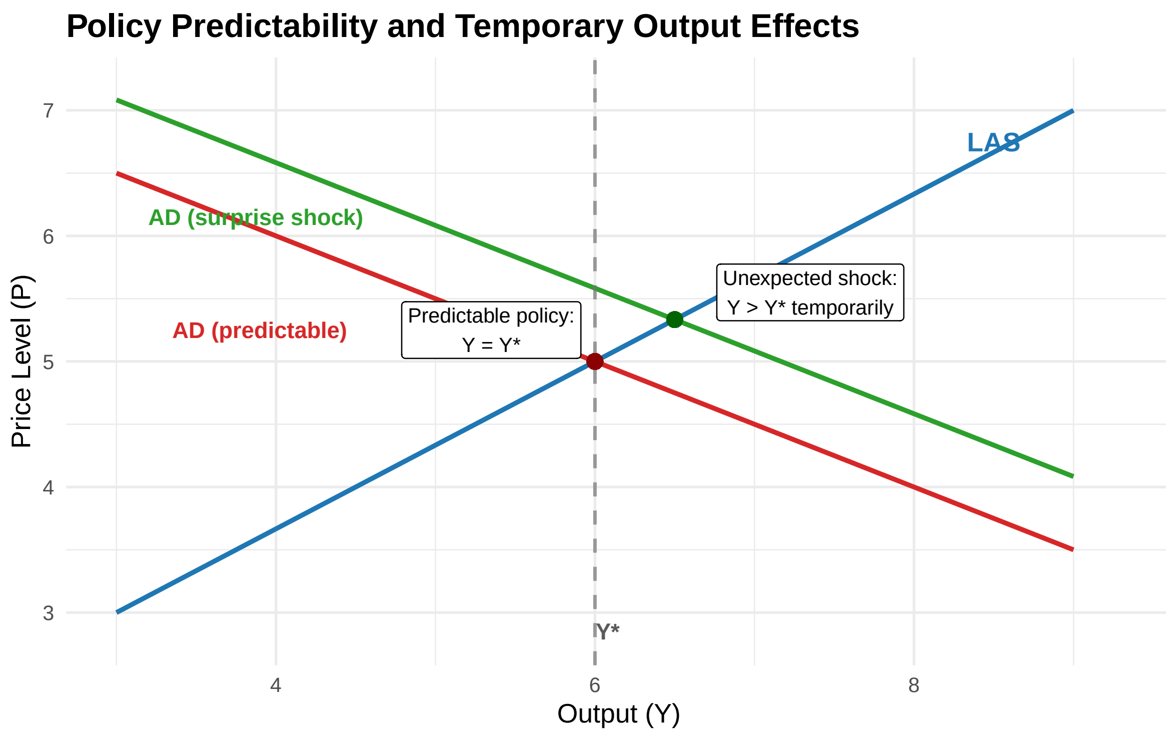

Figure 2: Predictable monetary policy leaves output at potential because the AD curve intersects Lucas aggregate supply at the benchmark point Y^*. An unexpected monetary expansion shifts AD rightward and temporarily raises output above Y^*.

where g_M is the systematic growth rate of money and u_t is a random monetary shock. The shock is white noise, so it has zero expected value conditional on information available in period $t-1$:

$$E_{t-1}u_t = 0.$$

Then:

$$E_{t-1}M_t = (1+g_M)M_{t-1},$$

because the predictable part of money growth is known one period in advance and only u_t is unforecastable. Therefore:

$$M_t - E_{t-1}M_t = u_t.$$

Output deviations:

$$Y_t - Y^* = \frac{\kappa}{1+\kappa}u_t.$$

Only randomness affects output. Predictable money growth affects the price level, not real activity.

3.3 Why deliberate unpredictability is not recommended

Even if monetary surprises can move output temporarily, that does not imply central banks should try to engineer surprise as a strategy. The New Classical argument runs in the opposite direction: frequent unpredictability degrades the information content of prices and makes supply less responsive.

The figure below illustrates that logic. Under a stable policy regime, firms treat price movements as relatively informative, so the Lucas aggregate supply curve is flatter. Under a volatile policy regime, firms discount price movements as noise, the effective value of κ falls, and the LAS curve becomes steeper.

The model should not justify randomness. Two reasons:

1. Random policies produce recessions and booms equally (negative u_t as likely as positive).
2. High volatility changes behaviour: agents discount price signals, steepening LAS (lower effective κ, higher price response).

This logic explains modern central-bank transparency: policy is more effective when expectations are anchored and communication reduces unnecessary uncertainty.

4 Exceptions and policy extensions

Two important exceptions: monetary policy can stabilise real activity under rational expectations if key assumptions are relaxed.

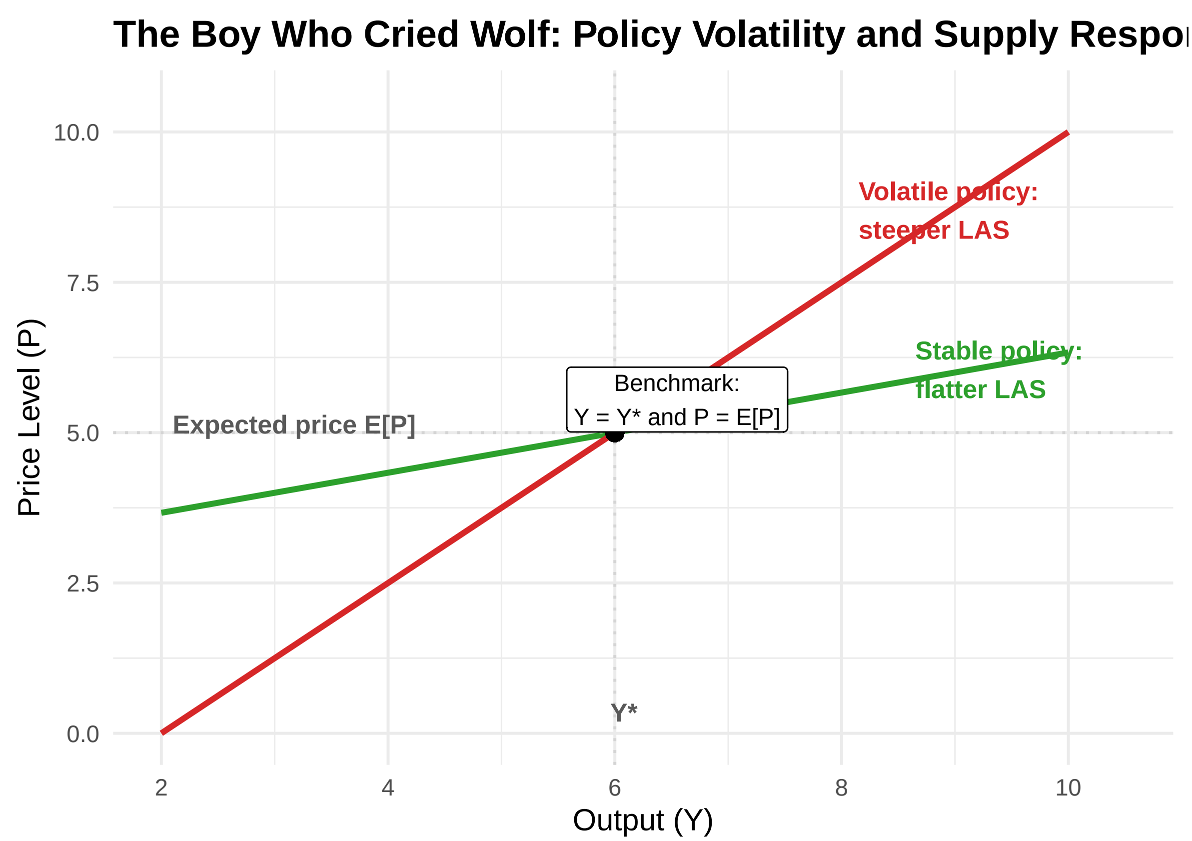

Figure 3: Higher policy volatility steepens Lucas aggregate supply through information effects. With stable policy, firms respond more to price signals and the LAS curve is flatter. With volatile policy, firms discount those signals, so the LAS curve becomes steeper.

4.1 Asymmetric information and real shocks

If productivity shocks drive cycles and the central bank forecasts a negative shock before the private sector, it can expand policy in advance to stabilise output (at higher prices).

This fits the New Classical framework because the CB has an informational advantage, creating a price surprise from the public's perspective.

Compatible with real-business-cycle reasoning: technology/productivity shocks drive cycles.

4.2 Long-term nominal wage contracts and stabilisation

In many economies, nominal wages are fixed by multi-year contracts. Even if demand shocks are anticipated, monetary policy can offset them because some nominal wages cannot adjust immediately—rigidity creates stabilisation scope.

This aligns with Fischer (1977): under wage rigidity, monetary rules interact with shock propagation even under rational expectations.

Under symmetric information (no CB informational advantage), stabilisation works best for demand shocks; supply-side stabilisation requires informational asymmetry.

4.3 Lucas critique and credibility

The **Lucas critique**: policy-effect estimates from historical relationships are unreliable if decision rules shift with the regime. This motivates structural, microfounded models.

If agents form expectations rationally, **commitment and credibility** shape outcomes through expectations. Rules, strategy statements, and transparent communication are policy tools.

Time-inconsistency literature: policy is strategic interaction with rational agents. Discretionary optimisation can backfire because promises today are not optimal tomorrow (after expectations set). Central to rules-versus-discretion analysis.

5 Modern evidence and applications

5.1 Measuring policy surprises

New Classical model disturbance: $M_t - E_{t-1}M_t$. Modern empirical work isolates surprise components using high-frequency financial data.

Recent IMF working paper: dataset of monetary policy shocks for 21 advanced and 8 emerging economies (2000–2022), using daily changes in interest rate swap rates around central bank announcements to identify unexpected policy shocks to the rate path. Maps directly to model: unexpected shocks are the empirical analogue of u_t.

5.2 Credibility and anchored expectations in practice

New Classical analysis makes credibility central. Modern CBs frame policy as expectations management.

Federal Reserve: publishing its "Statement on Longer-Run Goals and Monetary Policy Strategy" helps the public understand the framework—matters for transparency, accountability, and effectiveness.

ECB (2021 strategy): symmetric 2% medium-term inflation aim; notes that effective lower bound can require "especially forceful or persistent" measures, including transitory inflation above target.

IMF research: better-anchored expectations → less persistent inflation responses to shocks, reinforcing New Classical emphasis on credibility and expectations formation.

5.3 Policy regime episodes

Bank of Japan: reference timeline documents unconventional regimes from late 1990s onward (ZIRP 1999, QE from 2001). Explicitly lists "policy duration effect (forward guidance)" as part of ZIRP—an expectations-management concept compatible with New Classical theory.

Japan's 2024 framework change: QQE with yield curve control and negative rates judged to have "fulfilled their roles," shifting back to short-rate guidance—regime transition where expectations and credibility matter.

ECB (June 2014): introducing negative deposit facility rate (–0.10% effective June 11, 2014) illustrates how rules change when constraints bind and why expectation formation is central to transmission.

COVID-19 emergency actions: central banks reacted with large-scale measures for market functioning and macro stabilisation, showing regime and expectations logic when short rates are constrained.

5.4 Comparative framework: mechanisms and applications

Mechanism	Prediction	Modern illustration
Systematic, predictable policy	If anticipated, shifts $E_{t-1}P_t$ one-for-one; $Y_t \approx Y^*$	Fed and ECB strategy statements anchor expectations through transparency.
Unanticipated shocks	Only surprise component drives $P_t - E_{t-1}P_t$, hence $Y_t - Y^*$	IMF high-frequency dataset (2000–2022) identifies unexpected policy shocks at announcements.
Policy volatility	Agents discount price signals; LAS steepens (lower effective κ)	CBs prefer predictable reaction functions over crisis-era surprise moves.
Wage contracts	Nominal rigidity reopens stabilisation, even under rational expectations	BoE COVID interventions supported market functioning and prevented tightening.
Informational advantage	CB with better real-shock information can offset systematically	BoJ forward guidance and regime design link to RBC-style productivity shocks.

6 Problem questions

1. In the LAS equation $Y_t - Y^* = \kappa(P_t - E_{t-1}P_t)$, interpret every term economically. Why does the model predict $E_{t-1}Y_t = Y^*$ under rational expectations?
2. Show that equilibrium output deviations are $Y_t - Y^* = (\kappa/(1+\kappa))(M_t - E_{t-1}M_t)$. What happens as $\kappa \to 0$? What happens as $\kappa \to \infty$?
3. Explain why "random policy can move output" is not an argument for deliberate policy randomness. Explain using both (i) the sign symmetry of shocks and (ii) the "boy who cried wolf" mechanism.
4. Describe the asymmetric-information exception: what must be true about the central bank's information set relative to the private sector's for systematic stabilisation to operate under rational expectations?
5. Explain the wage-contract critique: why can anticipated monetary policy offset demand shocks when only part of the economy's wages are flexible?
6. Using high-frequency monetary policy shock data from modern empirical work, explain how researchers operationalise "policy surprises." Why is this relevant to the New Classical policy-ineffectiveness result?

6.1 References

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